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Potassium iodide

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Introduction

- **Potassium iodide (KI) is a medication and dietary supplement.**

- **As a dietary supplement:**
  - It has utility in patients with **low iodine intake**.

- **As a prescribed medication:**
  - It is used to treat:
    - Severe hyperthyroidism.
    - Cutaneous inflammatory dermatoses.
    - Nuclear emergencies.
    - Protect the thyroid gland when using radiopharmaceuticals.

Mechanism of Action

KI has several mechanisms of action on thyroid function

- **The most rapid effect (hours to days).**
  - Decreases thyroglobulin proteolysis, thereby decreasing thyroid hormone secretion.

- **Secondly, KI inhibits thyroid hormone synthesis.**
  - Temporary inhibition of iodine organification in the thyroid gland → decreasing thyroid hormone biosynthesis, a phenomenon called the **Wolff – Chaikoff effect**.

- **It is used in preparation for thyroidectomy because it decreases the vascularity of the thyroid gland.**
  - This decreases the risk of post-thyroidectomy hemorrhaging

Therapeutic Uses

Severe Hyperthyroidism Or Refractory Hyperthyroidism

- **Patients receive KI for short-term use in the following situations:**

1) Preoperative preparation for **thyroidectomy** in **Graves disease**.
2) The **thyroid storm** because iodine blocks the release of T4 and T3 from the gland within hours.
3) As an **adjunctive therapy** for Graves disease, used in combined treatment with antithyroid drugs.

Therapeutic Uses (Continue)

Cutaneous Inflammatory Dermatoses

- **The two best indications in this group are neutrophilic dermatoses and panniculitis.**
  - Especially for lymphocutaneous and cutaneous sporotrichosis:
  - Itraconazole is the drug of choice for the treatment.
  - However, patients who don't respond to itraconazole at 200 mg/day can receive KI together with other **antimycotics** as an alternative.

- **It is also successfully used for other inflammatory dermatoses.**
  - For instance, erythema nodosum, subacute nodular migratory panniculitis, nodular vasculitis, erythema multiforme, and sweet syndrome.

Therapeutic Uses (Continue)

Concerning Emergency Radiation

- **The U.S. Nuclear Regulatory Commission (NRC) and the American Thyroid Association (ATA) require states to consider including KI as a protective measure.**
  - FDA **prioritizes age**; primary factor determining risk for radioiodine-induced thyroid cancer after radiation exposure.
  - **Highest risk are:**
    - Infants.
    - Children.
    - Pregnant and nursing females.
  - ➢ Treat them at the lowest threshold of the predicted radioactive dose to the thyroid.

- **Any person over 18 years old and up to 40 years old:**
  - Should receive treatment at a slightly higher limit.

- **Anyone over 40 years old**:
  - Should have KI treatment **only** if the predicted exposure level is high enough to destroy the thyroid.

Administration

Diseases	Treatment and Dosage
Inflammatory dermatoses	▪ Begin with an oral dosage of 300 mg (approximately six drops of supersaturated potassium iodide (SSKI)) 3 times daily. ▪ Followed by weekly increases as tolerated.
Mycoses	▪ Beginning at 600 mg (approximately 12 drops of SSKI) orally three times each day. ▪ Often increased to 6 g (about 127 drops of SSKI) daily if tolerated.

- **Most are given orally, usually with juice or milk, to protect against gastrointestinal irritation.**

- **Several FDA-approved KI products exist, including:**
  - Tablets (65 and 130 mg).
  - Oral solutions (65 mg/mL).

- **There exist another two liquid presentations prescribed orally:**
  - SSKI with 35 to 50 mg of iodine per drop and KI with about 24 mg per drop.
  - Potassium iodide-iodine (**Lugol solution** [5 to 8 mg of iodine per drop]) is usually given orally with the recommended dosage of 3 to 5 drops three times daily.

Monitoring

- **If KI use is for more than one month, it is recommended to do a screening test of TSH to ensure that the patients are not in hypothyroidism.**
  - If iodide-induced hypothyroidism is detected, these changes are **reversible** by discontinuing the administration of KI.

- **According to the FDA's guidance.**
  - Thyroid function should be monitored in pregnant or breastfeeding women, neonates, and young infants if repeat doses are necessary following radioactive iodine exposure.

Contraindications

- **KI is contraindicated in patients who have thyroid disease or are using any drug that could alter thyroid function.**
  - Contraindications also include patients with an **allergy** to iodine.
  - Avoid with patients with chronic renal failure because of the presence of potassium.
  - Avoided in patients using potassium-sparing diuretics or angiotensin-converting-enzyme inhibitors to prevent hyperkalemia.

- **Immunocompromised patients such as:**
  - Patients with cancer, cirrhosis, AIDS, autoimmune diseases, or poorly managed diabetics, transplant patients, and those using corticosteroids should not use KI as it affects the immune system.

- **It should not be indicated in pregnant or nursing women.**
  - Because it causes neonatal hypothyroidism, thyromegaly, fetal airway obstruction, and prolonged labor.
  - Also, it is a pregnancy **category D drug**.

Adverse Effects

- **Adverse effects are unlikely when KI is used at low doses and for a short time (less than two weeks).**
  - The most common side effects are on the digestive system.
  - Thus, the recommendation is to take it with juice or milk to protect against gastrointestinal irritation.
  - However, significant side effects may occur when high doses are administered, especially for treating infectious skin disorders.

- **The acute side effects include:**
  - Diarrhea, Nausea, Vomiting, Stomach Pain.

- **Prolonged use can cause:**
  - Iodism or potassium toxicity.
  - **Iodism:** is an **iodide poisoning syndrome characterized by** soreness of the teeth and gums, severe headache, conjunctival hyperemia, lacrimation, blurred vision, rhinorrhea, and sialorrhea.

- **Systemic side effects of SSKI include:**
  - Urticaria, fever, eosinophilia, jaundice, pruritus, angioedema, and bronchospasm.
  - ✓ In this case, the treatment is high-dose corticosteroid therapy.

Thank You